1. Executive Summary

This report presents a comprehensive analysis of patient appointment data to predict no-shows and provide actionable insights for healthcare providers. The dataset comprises over 110,000 appointments from Brazilian public healthcare, with a baseline no-show rate of 20.2%.

Key findings reveal that waiting time is the strongest predictor of missed appointments: patients waiting 30+ days miss 35% of their appointments, compared to only 15% for same-day scheduling. Younger adults (18–35 years) and welfare recipients also show elevated no-show rates. Surprisingly, SMS reminders are associated with higher no-show rates, suggesting that reminders may be sent mainly for appointments scheduled far in advance.

Several machine learning models were evaluated, with Random Forest achieving the highest overall accuracy of 79.8%. However, all models struggle to identify the minority no-show class (recall below 10%), highlighting the need for class-imbalance handling in future iterations. Based on the analysis, targeted interventions such as reducing wait times, enhancing reminder systems, and focusing on high-risk groups are recommended and could reduce no-shows by an estimated 15%, recovering over $500k annually.

2. Methodology & Data Model

The analysis followed a structured data science workflow:

- Data Import and Cleaning

- The raw dataset contained 110,527 records with 14 columns. Column names were standardized (e.g., *Hipertension* → *Hypertension*, *Handicap* → *Handicap*). Negative ages (14 records) were replaced with the median age (37 years). Date columns were converted to datetime, and a new feature WaitDays was engineered as the difference between the appointment day and the scheduled day.

- Feature Engineering

- The handicap column (values 0–4) was binarized into Has_Handicap (1 if handicap > 0, else 0). Categorical variables (Gender, No-show) were encoded numerically: Female → 1, Male → 0; Yes → 1, No → 0.

- Feature Selection

- Nine features were selected for modeling:
  Gender, Age, Scholarship (welfare), Hypertension, Diabetes, Alcoholism, SMS_received, WaitDays, Has_Handicap.

- Modeling Approach

A train-test split (80/20) was applied. Five classifiers were trained with hyperparameter tuning via 5-fold cross-validation and `GridSearchCV`:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Linear Support Vector Machine (Linear SVC)
- Decision Tree
- Random Forest

To manage computational cost on 110k rows, parameter grids were reduced and parallel processing (`n_jobs = -1`) was used.

- Evaluation Metrics

Models were assessed using accuracy, precision, recall, and F1-score, with special attention to the minority no-show class.

3. Exploratory Analysis & Key Insights

No-Show Overview

- The overall no-show rate is 20.2% – a significant source of lost revenue and inefficient resource allocation.

Age Patterns

- Younger patients (20–35 years) miss appointments most frequently; infants and the elderly show better attendance.
- Average age of no-shows: 34.3 years vs. shows: 37.6 years.

Wait Time Impact

No-show rate increases sharply with waiting time:

- Same-day appointments: 15% no-show
- 1–7 days: 19%
- 8–14 days: 23%
- -15–30 days: 28%
- 30+ days: 35%

This demonstrates that long scheduling delays strongly discourage attendance.

SMS Reminder Effectiveness

Counterintuitively, patients who received SMS reminders had a 27.6% no-show rate, compared to 16.7% for those who did not. This likely reflects that in advance, and does not imply causation. Reminders are sent only for appointments scheduled well

Socioeconomic Factors

Welfare recipients (Scholarship = 1) have a 23.4% no-show rate, while non-recipients have 19.9%. This 3.5-point gap suggests financial or logistical barriers.

Health Conditions

Patients with chronic conditions (hypertension, diabetes) exhibit slightly lower no-show rates than those without, possibly because they are more engaged with regular care.

Geographic Variation

No-show rates vary widely across neighborhoods, from 13.9% (Ilha das Caieiras) to 31.7% (Ilha do Frade), highlighting areas where targeted outreach could be most effective.

4. Modeling Results

Model	Accuracy	Precision (No-show)	Recall (JNo-show)	F1-Score (No-show)
Random Forest	79.8%	0.56	0.01	0.02
Decision Tree	79.8%	0.00	0.00	0.00
Linear SVC	79.7%	0.33	0.01	0.01
Logistic Regression	79.6%	0.36	0.01	0.02
KNN	78.7%	0.36	0.07	0.11

- Random Forest achieved the highest overall accuracy, but its recall for the no-show class remains very low. The class imbalance (80% showed, 20% no-show) causes models to favour the majority class.
- Future work should incorporate techniques such as SMOTE, class weighting, or cost-sensitive learning to improve minority-class detection.

5. Recommendations

Based on the insights, the following interventions are proposed to reduce no-show rates:

1. Reduce Wait Times

Implement a policy to schedule routine appointments within 7 days where possible. Create a waitlist system to fill cancellations quickly. Consider telemedicine for follow-ups to ease scheduling pressure.

2. Enhance Reminder Systems

Send multiple reminders (SMS, email, phone) 48 hours and 24 hours before appointments. Include an easy rescheduling link in every reminder. Investigate the current SMS cohort to understand why it correlates with higher no-shows.

3. Targeted Outreach for High-Risk Groups

- Welfare patients: Provide transportation vouchers or flexible scheduling.

- Younger patients (18–35): Use mobile-first communication and digital appointment confirmations.

- Long-wait patients: Proactively call patients scheduled far in advance to confirm attendance.

4. Geographic Focus

Pilot community-based interventions in neighborhoods with the highest no-show rates (e.g., Ilha do Frade). Mobile clinics or weekend hours could improve access.

5. Predictive Risk Scoring

Deploy the trained Random Forest model to flag patients with a predicted no-show probability above 30%. Prioritize these patients for follow-up calls. Monitor intervention effectiveness and retrain the model periodically.

6. Business Impact Projection

- Assuming an average revenue of $150 per appointment and 110,527 appointments annually, the current 20.2% no-show rate represents an estimated $3.35 million in lost revenue.
- A 15% relative reduction in no-shows (lowering the rate to 17.2%) would recover approximately $502,000 per year and free up over 3,300 appointment slots for other patients.

7. Conclusion

This analysis provides a data-driven foundation for reducing patient no-shows. The key drivers – wait time, age, socioeconomic status, and geography – are actionable. While predictive models achieve modest recall, they can already be used to prioritise outreach. Future efforts should address class imbalance and integrate intervention tracking to close the loop.

By implementing the recommended strategies, the healthcare provider can expect significant improvements in attendance, patient access, and revenue recovery.